Digital Lock-In Amplifier

Design and Implementation using STM32H743

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Motivation

- Many scientific experiments require detection of extremely weak signals.
- These signals are often buried in large noise sources.
- Conventional instruments struggle to isolate such signals.
- Lock-in amplifiers enable extraction of signals at a specific frequency.
- Commercial lock-in amplifiers are expensive.

Project Goal

- Design a low-cost digital lock-in amplifier.
- Implement synchronous detection using STM32.
- Develop a custom mixed-signal PCB.

What is a Lock-In Amplifier

- A lock-in amplifier extracts signals at a specific frequency.
- It rejects noise outside the reference frequency.
- Uses synchronous detection technique.
- Enables measurement of signals smaller than the noise level.

Applications:

- Optical sensing
- Magnetic measurements
- Biomedical instrumentation
- Precision laboratory experiments

Principle of Lock-In Detection

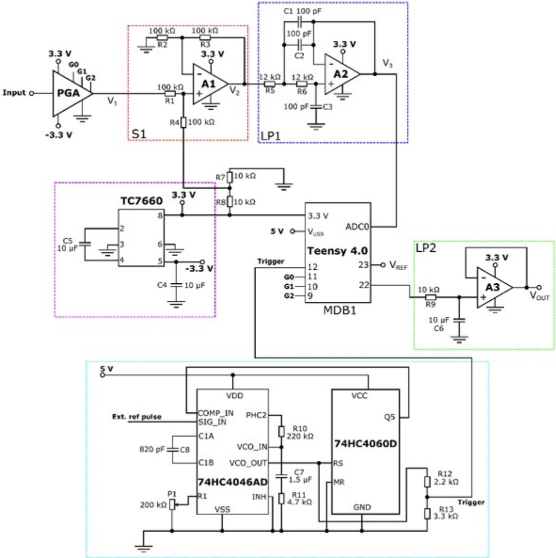
- Input signal contains both desired signal and noise.
- Signal is multiplied with a reference waveform.
- Desired signal shifts to DC after multiplication.
- Low-pass filtering removes unwanted components.

$$V_{sig} = A_{sig} \cos(\omega t + \theta)$$

$$V_{ref} = A_{ref} \cos(\omega t)$$

$$Output = \frac{A_{sig} A_{ref}}{2} \cos(\theta)$$

Reference Design: OpenLockin



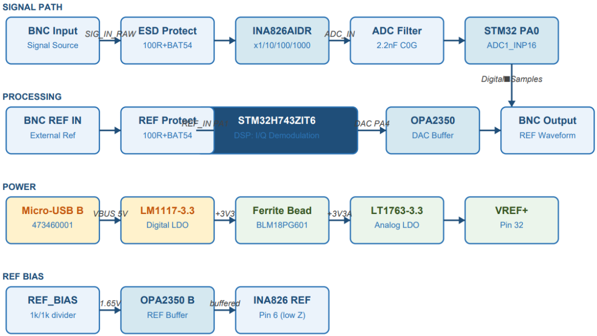
Our Design Improvements

- Replace Teensy controller with STM32H743 microcontroller.
- STM32 operates at **480 MHz**.
- Integrated ADC simplifies signal acquisition.
- Custom mixed-signal PCB design.
- Improved analog front-end using instrumentation amplifier.

Advantages:

- Higher computational performance
- Integrated DSP capability
- Scalable system architecture

System Block Diagram



Signal processing flow:

- Signal input through BNC connector
- Analog signal conditioning
- ADC sampling using STM32
- Digital lock-in algorithm
- Output signal generation

Signal Flow in the System

1. Signal enters through BNC input connector.
2. Instrumentation amplifier increases signal amplitude.
3. Offset added using REF pin (1.65V midpoint).
4. RC anti-alias filter suppresses high-frequency noise.
5. STM32 ADC samples the signal.
6. Digital lock-in algorithm extracts amplitude and phase.

Analog Front-End

Instrumentation amplifier used:
INA826

- Amplifies weak input signals before ADC sampling.
- Gain selectable from $\times 10$ to $\times 1000$.
- High common-mode rejection ratio.
- Low-noise performance.

Additional features:

- Input protection using Schottky diodes.
- REF pin used to shift signal to mid-supply.
- RC anti-alias filter before ADC.

Main Controller

STM32H743 Microcontroller

- ARM Cortex-M7 processor
- Operating frequency up to 480 MHz
- High-speed ADC for signal acquisition
- DMA support for efficient data transfer
- Hardware timers for precise synchronization

Responsibilities:

- Signal acquisition
- Reference waveform generation
- Digital demodulation
- Communication with host system

Digital Lock-In Algorithm

Processing steps implemented in firmware:

1. Generate sine and cosine reference signals.
2. Sample input signal using ADC.
3. Multiply signal with reference signals.
4. Apply digital low-pass filtering.
5. Compute amplitude and phase.

$$R = \sqrt{I^2 + Q^2}$$

$$\theta = \tan^{-1}\left(\frac{Q}{I}\right)$$

Power Architecture

Low-noise power design is essential for accurate measurements.

- USB powered system (5V input).
- Low-noise LDO regulator generates 3.3V rail.
- Ferrite bead isolation between analog and digital supplies.
- Separate AGND and DGND domains.
- Star-ground connection near MCU.

PCB Stackup

4-layer PCB used for mixed-signal design.

- Layer 1: Top signal layer and components
- Layer 2: Continuous ground plane
- Layer 3: Power plane (3.3V)
- Layer 4: Bottom routing layer

Advantages:

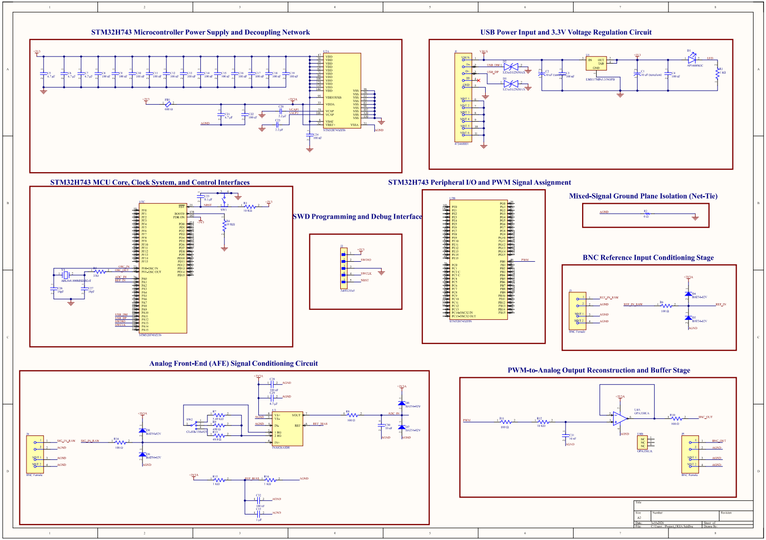
- Low ground impedance
- Reduced EMI
- Improved ADC performance

PCB Design Considerations

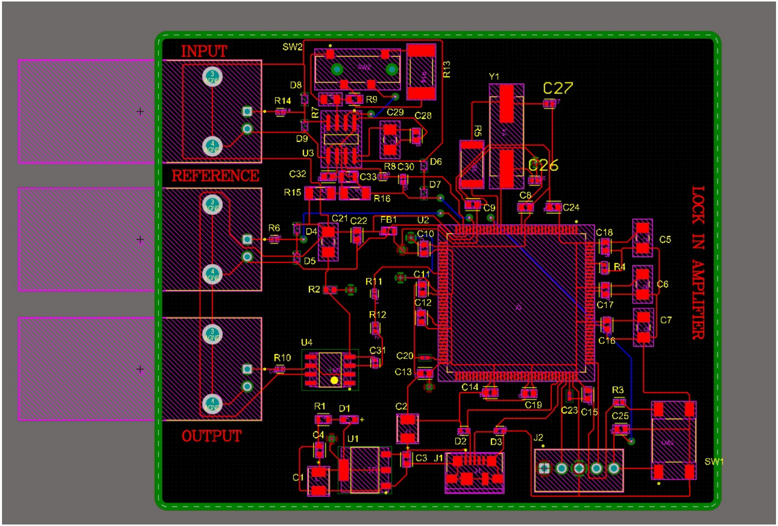
Key layout features:

- Instrumentation amplifier placed close to BNC input
- Short analog signal paths
- Continuous ground plane
- Proper decoupling capacitors near MCU
- Separation of analog and digital sections
- Controlled impedance routing for USB

Schematic


STM32H743 Microcontroller Power Supply and Decoupling Network
USB Power Input and 3.3V Voltage Regulation Circuit
STM32H743 MCU Core, Clock System, and Control Interfaces
STM32H743 Peripheral I/O and PWM Signal Assignment
Mixed-Signal Ground Plane Isolation (Net-Tie)
SWD Programming and Debug Interface
BNC Reference Input Conditioning Stage
Analog Front-End (AFE) Signal Conditioning Circuit
PWM-to-Analog Output Reconstruction and Buffer Stage

PCB Layout


INPUT
REFERENCE
OUTPUT
LOCK IN AMPLIFIER
SW2
R13
Y1
C27
D8
R14
R9
R7
C29
C28
D9
R5
U3
C26
R8
C30
D6
C32
C33
D7
R15
R16
C8
C9
C24
R6
C21
C22
FB1
U2
D4
C10
C18
C5
D5
R4
R2
R11
C11
C6
C12
C17
R12
C16
C7
U4
C31
R10
C20
C13
C14
C19
R3
R1
D1
C2
C23
C15
C25
C4
U1
D2
D3
J2
C3
J1
SW1
C1

Expected Performance

- Frequency range: 10 Hz – 50 kHz
- Gain range: $\times 10$ – $\times 1000$
- ADC resolution: 12-bit
- Sampling rate: up to 1 MSPS
- Detection of microvolt-level signals
- USB-powered portable system

Future Work

- PCB fabrication and assembly
- Hardware validation and calibration
- Noise floor characterization
- Advanced digital filtering techniques
- PC interface for data visualization

Conclusion

- Designed a digital lock-in amplifier architecture.
- Integrated analog signal conditioning with STM32 DSP.
- Developed a custom mixed-signal PCB.
- Provides a low-cost alternative to commercial lock-in amplifiers.

Thank You